The Exact Spectral Gap of Kac's Master Equation: A Complete Conditional-Projection Induction

HCS-C322 theorem-and-evidence package

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Abstract

We give a normalization-locked proof of the exact gap for Kac's uniform binary-collision master equation. On $S^{N-1}(\sqrt{N})$, with unordered-pair angle average Q_N and positive generator $L_N = N(I - Q_N)$, the gap is $(N+2)/[2(N-1)]$. We reproduce the infinite-dimensional lower induction via coordinate conditional expectations, identify the quartic equality mode and its multiplicity, and close the $N = 2$, positive-energy, and zero-energy boundaries. Exact polynomial matrices audit the algebra but are not offered as the lower-bound proof.

1 Frozen collision operator

Let σ_N be normalized surface measure on $X_N = S^{N-1}(\sqrt{N})$, $N \geq 2$. If $R_{ij}(\theta)$ rotates the (i, j) coordinate plane, define

$$Q_N f(v) = \binom{N}{2}^{-1} \sum_{i<j} \frac{1}{2\pi} \int_{-\pi}^{\pi} f(R_{ij}(\theta)v)\, d\theta, \qquad L_N = N(I - Q_N). \tag{1}$$

Thus pairs are unordered, the angular measure is a probability, and the factor N belongs to the generator. Kac introduced this collision model in kinetic theory [1]; the exact gap architecture below is due to Carlen, Carvalho, and Loss [2].

Each pair average is the orthogonal projection onto the functions invariant under its circle action. Hence Q_N is a positive self-adjoint Markov contraction. Its fixed functions are constants: coordinate-plane rotations generate $SO(N)$, which is transitive on X_N. Put

$$\lambda_N = \sup_{\substack{\|f\|_2=1 \\ \langle f,1\rangle=0}} \langle f, Q_N f\rangle, \qquad \Delta_N = N(1 - \lambda_N).$$

Let

$$F_N(v) = \sum_{i=1}^{N} v_i^4 - \frac{3N^2}{N+2}.$$

The centering uses $\mathbb{E}_{\sigma_N} v_i^4 = 3N/(N+2)$. Direct angle averaging gives

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \{(u\cos\theta + v\sin\theta)^4 + (-u\sin\theta + v\cos\theta)^4\}\, d\theta = \frac{3}{4}(u^2 + v^2)^2. \tag{2}$$

Summing (2) over pairs and using $\sum_i v_i^2 = N$ yields

$$Q_N F_N = \left(1 - \frac{N+2}{2N(N-1)}\right) F_N, \qquad L_N F_N = \frac{N+2}{2(N-1)} F_N. \tag{3}$$

This is an upper bound for the gap, not yet a proof that no slower mode exists.

2 The conditional-projection lower bound

For $P_j f = \mathbb{E}(f \mid v_j)$ set $P = N^{-1} \sum_j P_j$. Fixing $v_j = y$ leaves an $(N-1)$-coordinate sphere. Let $Q_{N-1}^{(j)}$ be the pair average excluding j, with the remaining sphere rescaled if needed. Counting each pair in exactly $N-2$ slices gives

$$\langle f, Q_N f\rangle = \frac{1}{N} \sum_{j=1}^{N} \int \langle f_{j,y}, Q_{N-1}^{(j)} f_{j,y}\rangle\, d\nu_N(y). \tag{4}$$

Equivalently, if $L_{N-1}^{(j)} = (N-1)(I - Q_{N-1}^{(j)})$, then $L_N = (N-1)^{-1} \sum_j L_{N-1}^{(j)}$ as quadratic forms. Centering every slice by $P_j f(y)$ in (4) and applying λ_{N-1} gives, for unit mean-zero f,

$$\langle f, Q_N f\rangle \leq \lambda_{N-1} + (1 - \lambda_{N-1})\langle f, Pf\rangle. \tag{5}$$

The exact norm of P is therefore decisive.

Lemma 1 (Coordinate-projection spectrum)**.** *For $N \geq 3$, on constants-perpendicular $L^2(\sigma_N)$,*

$$\|P\| = \mu_N = \frac{N+4}{N(N+1)}.$$

Its top eigenspace is one-dimensional and is generated by F_N.

Proof. On one-coordinate functions define the self-adjoint conditional operator $(K_N g)(x) = \mathbb{E}[g(v_2) \mid v_1 = x]$. Conditional spherical moments give

$$K_N(x^{2r}) = c_{r,N}(N - x^2)^r, \quad c_{r,N} = \frac{(2r-1)!!}{(N-1)(N+1)\cdots(N+2r-3)}, \tag{6}$$

and symmetry kills odd functions. Even polynomial spaces are invariant. Self-adjointness and density of polynomials for the compact coordinate marginal produce a complete orthogonal-polynomial eigenbasis, whose eigenvalues are the leading coefficients in (6):

$$\alpha_0 = 1, \qquad \alpha_{2r} = (-1)^r c_{r,N}, \qquad \alpha_{2r+1} = 0.$$

Since $|\alpha_{2r+2}/\alpha_{2r}| = (2r+1)/(N+2r-1) < 1$, the largest mean-zero eigenvalue is $\kappa_N = \alpha_4 = 3/(N^2-1)$ and the smallest is $\alpha_2 = -1/(N-1)$.

We now transfer this spectrum without presuming that P has only point spectrum. Put $H_0 = L_0^2(\nu_N)$ and define $T : H_0^N \to L_0^2(\sigma_N)$ by $T(h_1, \dots, h_N) = \sum_j h_j(v_j)$. Then $T^* f = (P_1 f, \dots, P_N f)$ and $P = TT^*/N$. The nonzero spectra of TT^*/N and T^*T/N agree: besides the usual eigenvector transfer, for $z \neq 0$ the resolvent identity

$$(TT^* - z)^{-1} = -z^{-1} I + z^{-1} T(T^*T - z)^{-1} T^*$$

(and its adjoint counterpart) rules out unaccounted continuous or approximate spectrum. The block matrix of T^*T/N has diagonal I/N and off-diagonal K_N/N. On a K_N eigenfunction of eigenvalue α, the trivial and standard coordinate-index subspaces therefore have eigenvalues

$$\frac{1 + (N-1)\alpha}{N}, \qquad \frac{1 - \alpha}{N}.$$

The complete polynomial resolution above makes these blocks exhaustive; their only accumulation is $1/N$, already the infinite-multiplicity odd block. Any additional point for P is zero from $\ker T^*$. Consequently the largest mean-zero spectral value is

$$\max\left\{\frac{1 + (N-1)\kappa_N}{N}, \frac{1 - \alpha_2}{N}\right\} = \frac{N+4}{N(N+1)}.$$

The second entry is $1/(N-1)$, strictly smaller for $N > 2$. Simplicity of α_4 and of the trivial index line, followed by T, identifies the unique top vector with the sum of centered coordinate quartics, namely F_N. This proves both the norm and uniqueness. $\square$

Theorem 1 (Exact gap, multiplicity, and decay). *For every $N \geq 2$,*

$$\mathrm{gap}(L_N) = \Delta_N = \frac{N+2}{2(N-1)}.$$

For $N \geq 3$ the slow eigenspace is $\mathrm{span}\{F_N\}$. *At $N = 2$, every mean-zero function is a slow eigenfunction. For every mean-zero f and $t \geq 0$,*

$$\|e^{-tL_N} f\|_2 \leq e^{-t\Delta_N} \|f\|_2,$$

and the rate is sharp.

Proof. Taking the supremum in (5) and applying Lemma 1,

$$\begin{aligned}\Delta_N &\geq \frac{N}{N-1}(1 - \mu_N)\Delta_{N-1} \\ &= \left(1 - \frac{3}{N^2-1}\right)\Delta_{N-1} = \frac{(N-2)(N+2)}{(N-1)(N+1)}\Delta_{N-1}.\end{aligned}$$

For $N = 2$, the sole pair is averaged over the entire circle, so Q_2 is projection onto constants and $\Delta_2 = 2$. Therefore

$$\Delta_N \geq 2\prod_{j=3}^{N} \frac{(j-2)(j+2)}{(j-1)(j+1)} = 2\,\frac{N+2}{4(N-1)} = \frac{N+2}{2(N-1)}.$$

Equation (3) attains this lower bound. For $N \geq 3$, equality in the induction forces equality in the P variational bound; Lemma 1 then gives the one-dimensional slow eigenspace. The $N = 2$ multiplicity follows from $Q_2 = 0$ on the whole mean-zero sector. The spectral theorem for the nonnegative self-adjoint L_N gives the semigroup bound, and the stated eigenspaces make it sharp. $\square$

Energy boundary. On $S^{N-1}(\sqrt{E})$ for any $E > 0$, dilation by $\sqrt{E/N}$ unitarily conjugates the walk above and leaves the gap unchanged; the centered quartic is $\sum_i v_i^4 - 3E^2/(N+2)$, since $\mathbb{E}v_i^4 = 3E^2/[N(N+2)]$. At $E = 0$ the state space is one point and the mean-zero Hilbert space is $\{0\}$, so no positive relaxation gap is assigned.

3 Exact evidence, ownership, and Route A

Exact rational evidence computes the degree-zero through degree-eight conditional polynomials for $3 \le N \le 12$, all gap-recursion rows for $2 \le N \le 12$, and Gram and Q forms on even polynomial test spaces for $2 \le N \le 7$. A producer-independent expansion reconstructs 6,540 form cells and checks self-adjointness. Symbolic, replay, strict-parser, repaired-hash mutation, optimized-mode, font, raster, and fresh-build gates are separate. These finite matrices audit algebra; the infinite-dimensional lower bound is Lemma 1 plus Theorem 1, not computation.

The Kac ring in C170 is deterministic, C183 acts on permutations, and C313 treats deterministic sphere geodesics; none is this continuous-sphere random collision operator. We make no claim about the full spectrum, entropy production, nonlinear Boltzmann convergence, nonuniform angles, or literature priority.

The Route-A tuple is $(A0, A1, A2, A3, A4) = (\mathrm{FAIL}, \mathrm{FAIL}, \mathrm{FAIL}, \mathrm{FAIL}, \mathrm{FORMAL\ HINT})$. The canonical self-adjoint Markov generator supplies the hint, but it is not a unitary/scattering/Hamiltonian quantization and has no arithmetic carrier or orbit-zeta bridge. Thus the verdict is `ROUTE_A_REJECTED` and Route B is locked. Under `NO_BAD_EULER_OR_ROOT_NUMBER`, no target arithmetic local data, Euler factor, root number, automorphy, target divisor, functional equation, zero match, or Hilbert–Pólya operator is asserted.

Revision certificate: independent polynomial evidence and scope closure

This final round adds exact independent reconstruction, source and collision ownership, hostile gates, and the complete Route-A firewall.

References

[1] M. Kac, "Foundations of kinetic theory," in *Proc. Third Berkeley Symp. Math. Statist. Probab.*, vol. III (1956), 171–197, UC Berkeley Library record 112857.

[2] E. A. Carlen, M. C. Carvalho, and M. Loss, "Determination of the spectral gap for Kac's master equation and related stochastic evolution," *Acta Math.* 191 (2003), DOI: 10.1007/BF02392695; arXiv:math-ph/0109003.